

1. Obesity is widely regarded as a pandemic, with potentially disastrous consequences for human health.
2. In developing countries, average national rates of obesity are low, but these figures may disguise: high rates of obesity in urban communities; for example, nearly one-quarter of women in urban India are overweight.
3. There is increasing public awareness of the health implications of obesity.
4. Many patients will seek medical help for their obesity, others will present with one of the complications of obesity, and increasing numbers are being identified during health screening examinations.
5. It is suggested that obesity at age 40 years can reduce life expectancy by up to 7 years for non-smokers and by 13 years for smokers.
6. Coronary heart disease is the major cause of death but cancer rates are also increased in the overweight, especially colorectal cancer in males and cancer of the gallbladder, biliary tract, breast, endometrial and cervix in females.
7. Obesity has little effect on life expectancy at > 70 years, but the obese do spend a greater proportion of their active life disabled.
8. Epidemic obesity has been accompanied by epidemic of type 2 diabetes and osteoarthritis, particularly of the knee.
9. Obesity may have profound psychological consequences, compounded by stigmatisation of the obese in many societies.

Complications of obesity

Obesity has adverse effects on both mortality and morbidity :

Risk factors	Outcomes
1. Metabolic syndrome Type 2 diabetes Hypertension	Coronary heart disease Stroke

Hyperlipidaemia	Diabetes complications
2. Liver fat accumulation Cirrhosis	Non-alcoholic steatohepatitis
3. Restricted ventilation Sleep apnoea Respiratory failure (Pickwickian syndrome)	Exertional dyspnoea
4. Mechanical effects of weight	Urinary incontinence Osteoarthritis Varicose veins
5. Increased peripheral steroid interconversion in adipose tissue	Hormone-dependent cancers (breast, uterus) Polycystic ovary syndrome (infertility, hirsutism)
6. Others	Psychological morbidity (low self-esteem, depression) Socioeconomic disadvantage (lower income, less likely to be promoted) Gallstones Colorectal cancer Skin infections (groin and sub mammary candidiasis; hidradenitis)

Body fat distribution

1. Increased intra-abdominal fat causes central ('abdominal', visceral', 'android' or 'appleshaped') obesity and it is more common in men and is more closely associated with type 2 diabetes, the metabolic syndrome and cardiovascular disease.
2. Increased subcutaneous fat accumulation causing 'generalised' ('gynoid' or 'pearshaped') obesity

Aetiology

Accumulation of fat results from a discrepancy between energy consumption and energy expenditure which is too large to be defended by the hypothalamic regulation of BMR.

Some reasons for the increasing prevalence of obesity-the 'obesogenic' environment

1. Increasing energy intake
 - Portion sizes
 - Energy-dense food
 - Snacking and loss of (mainly fat) regular meals
 - ↓ Affluence
2. Decreasing energy expenditure
 - Car ownership
 - ↓ Sports in schools
 - ↓ Walking to school/work
 - Time spent on computer games and watching TV
 - ↓ Manual labour
 - Automation;
 - Central heating

Potentially reversible causes of weight gain :

1. Endocrine factors
 - Hypothyroidism
 - Hypothalamic tumours or injury
 - Cushing's syndrome
 - Insulinoma
2. Drug treatments
 - Tricyclic antidepressants
 - Corticosteroids
 - Sulphonylureas
 - Sodium valproate
 - Oestrogen-containing
 - Contraceptive pill

Susceptibility to obesity

1. Susceptibility to obesity and its adverse consequences undoubtedly varies between individuals.
2. It is not true that obese subjects have a 'slow metabolism', since their BMR is higher than that of lean subjects.
3. Twin and adoption studies confirm a genetic

influence on obesity.

Reversible causes of obesity and weight gain

These patients are distinguished from those with idiopathic obesity by their short history, with a recent marked change in the trajectory of their adult weight

Clinical assessment and investigations

1. Severity of obesity can be quantified using the BMI

Quantifying obesity with body mass index (weight/height²)

Risk of obesity

BMI (kg/m ²)	Classification*	Comorbidity
18.5-24.9	Normal range	Negligible
25.0-29.9	Overweight	Mildly increased
> 30.0	Obese	
30.0-34.9	Class I	Moderate
35.0-39.9	Class II	Severe
> 40.0	Class III	Very severe

2. A waist circumference of > 102cm in men or > 88cm in women indicates that the risk of metabolic and cardiovascular complications of obesity is high.
3. A dietary history may be helpful in guiding dietary advice, but is notoriously susceptible to under-reporting of food consumption.
4. It is important to consider 'pathological eating behaviour' (such as binge eating, nocturnal eating or bulimia, which may be the most important issue to address in some patients.
5. Alcohol is an important source of energy intake and should be considered in detail.
6. The history of weight gain may help diagnose underlying causes, such as hypothyroidism or Cushing's syndrome.
7. Obese patients should have thyroid function tests performed on one occasion
8. An overnight dexamethasone suppression test or 24-hour urine free cortisol if syndrome is suspected.

9. Monogenic and 'syndromic' causes of obesity are usually only relevant in children presenting with severe obesity.
10. The impact of obesity on the patient's life and work is a major consideration. Assessment of other cardiovascular risk factor is important.
11. Blood pressure should be measured with a large cuff, if required.
12. blood glucose and a serum lipid profile
13. Elevated serum transaminases occur in patients with non-alcoholic fatty liver disease.

Differential diagnosis of obesity

- **Is there associated hyperphagia** : If the patient recognizes that he or she has a ravenous appetite or eats more than is necessary, the possibility of an insulinoma or Frohlich's syndrome should be considered.
- **Is the obesity centripetal**: The presence of centripetal obesity, especially with moon facies, should suggest Cushing's syndrome.
- **Is the obesity mainly of the lower extremities**: This finding would suggest lipodystrophy.
- **Is there mental retardation**: The presence of mental retardation should suggest LaurenceMoonBardetBiedl syndrome.
- **What is the sex of the patient**: In male patients, one should consider Klinefelter's syndrome, and in female patients, one should consider polycystic ovary.
- **What drugs is the patient on** : Many drugs may cause obesity, most notably the tricyclic antidepressants and corticosteroids.

Diagnostic Workup

- Routine laboratory tests include a CBC, urinalysis, chemistry panel, 2-hour postprandial blood sugar, and thyroid profile.
- If an insulinoma is strongly suspected, plasma

insulin, C-peptide, a 24- to 36-hour fast, a 5-hour glucose tolerance test, and tolbutamide tolerance test may be done.

- If Cushing's syndrome is suspected, a 24-hour urine free cortisol and cortisol suppression test should be done.
- Pelvic ultrasound will help diagnose polycystic ovaries.
- Chromosomal analysis will help diagnose Klinefelter's syndrome.
- Perhaps a psychiatrist should be consulted.

MCQs

1. The formula for the body mass index is _____.
 - a. A person's weight in kilograms squared divided by the square of his/her height in meters (kg^2/m^2)
 - b. A person's weight in kilograms squared divided by his/her height in meters (kg^2/m)
 - c. A person's weight in kilograms divided by his/her height in meters (kg/m)
 - d. A person's weight in kilograms divided by the square of his/her height in meters (kg/m^2)
2. Which among the following are the causes of weight gain
 - a. Cushing's syndrome, Hypothyroidism & Corticosteroids
 - b. Hyperthyroidism
 - c. Diabetes Mellitus Type 2
 - d. Addison's disease
3. Centripetal obesity with moon facies, is seen in
 - a. Insulinoma
 - b. Cushing's syndrome
 - c. Laurence-Moon-Bardet-Biedl syndrome
 - d. Hypothyroidism

Answer

1.	d	2.	a	3.	b
----	---	----	---	----	---